COUNTRY PROFILES

Nuclear Power in Estonia

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- Almost half of Estonia's electricity comes from oil shale, which the country aims to phase out.
- In recent years, Estonia has made significant progress towards introducing nuclear energy.
- Fermi Energia is proposing to construct two GE Hitachi BWRX-300 reactors by 2035 at one of two sites currently under consideration.

Privacy - Terms

Electricity sector

Total generation (in 2023): 5.7 TWh

Generation mix: coal* 2.8 TWh (49%); biofuels & waste 1.5 TWh (26%); solar 0.7 TWh (13%); wind 0.7 TWh (12%).

Import/export balance: net import 3.3 TWh (imports 7.7 TWh; exports 4.4 TWh)

Total consumption: 6.0 TWh

Per capita consumption: c. 4400 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

** Estonia is unique in its reliance on domestic oil shale, an energy-rich sedimentary rock, which some data sources treat as a coal equivalent.*

Estonia completed the 350 MW DC cable interconnector with Finland, Estlink 1, in 2007. The 650 MW Estlink 2 followed in 2014. Estlink 3 is proposed to operate from 2035 with a capacity of 700 MW.

Energy policy

Estonia aims to achieve carbon neutrality by 2050. Accordingly, the country has reduced the use of domestic oil shale in the electricity sector by half since 2007, achieving significant reductions in greenhouse gas emissions. At the same time, energy intensity in manufacturing has declined steadily, with electricity use by industry dropping by more than half. The country has diversified its electricity sources by introducing biofuels, wind and solar, but turning away from domestic oil shale tipped Estonia from being a net energy exporter to an importer in 2018. To further advance decarbonization and to maintain security of supply, the country is working to introduce nuclear energy.

The system operator Elering anticipates that electricity consumption will reach 11.3 TWh by 2035 (from 6.0 TWh in 2023).

Nuclear developments

After the 2009 shutdown of the regionally significant Ignalina nuclear power plant in Lithuania, Estonia had been part of an agreement with Poland, Latvia and Lithuania aiming to build a new nuclear plant at the same site. However, the project was discontinued in 2016.

In April 2021, the Estonian government formally approved the formation of a nuclear energy working group responsible for considering the possibility of introducing nuclear energy in Estonia. Environment Minister Tõnis Mölder, head of the group, stated: "In order to increase Estonia's energy security, sustainability and competitiveness and achieve the 2050 climate targets, the introduction of nuclear energy would be one of the possible solutions."

In January 2022, the USA and Estonia agreed to cooperate under the US Department of State's Foundational Infrastructure for Responsible Use of Small Modular Reactor Technology capacity-building programme.

In January 2023 a spatial analysis working group of the Estonian Ministry of Finance published an interim report proposing 16 possible coastal locations for nuclear power plants.

In May 2023, Estonia participated in the 'Nuclear Alliance', where France's Ministry for Energy Transition brought together counterparts from other EU states wishing to rely on nuclear energy, alongside renewables, to carry out their energy transition.

At the request of the Estonian government, the International Atomic Energy Agency conducted an eight-day Integrated Nuclear Infrastructure Review (INIR) mission in 2023, concluding that Estonia had developed a comprehensive assessment of its nuclear power infrastructure.

Members of the Riigikogu (parliament) submitted a draft resolution in May 2024 allowing preparations to begin for the adoption of nuclear energy, and the creation of a suitable legislative and regulatory framework. The following month, the Riigikogu passed a resolution (*Supporting the adoption of nuclear energy in Estonia (431 OE)*), stating that Estonia's national development plan up to

2035 must consider the impacts of adopting nuclear energy in order to ensure security of energy supply.

In January 2025 the International Atomic Energy Agency, in cooperation with Estonia's Ministry of Climate, conducted a legislative assistance mission to support the country in developing a legal framework for nuclear energy.

In December 2025, the Ministry of Climate submitted the draft Nuclear Energy and Safety Act (TEOS). Following public consultation the bill is expected to enter into force in January 2027. The legislation creates a legal framework for nuclear energy and establishes an independent nuclear regulator within the Consumer Protection and Technical Supervision Authority (TTJA).

Nuclear industry

In July 2019, Fermi Energia launched a feasibility study on the suitability of SMRs for Estonia. It selected four SMR designs to include: Moltex Energy's SSR-W300; Terrestrial Energy's IMSR-400; GE Hitachi's BWRX-300; and NuScale's VOYGR. Fermi Energia announced the selection of GE Hitachi's BWRX-300 as the winning bidder in February 2023.

In early 2020, Fermi Energia signed a memorandum of understanding with Finnish power company Fortum and Belgian engineering firm Tractebel to cooperate on studying the deployment of SMRs in Estonia. In June 2021, Swedish utility Vattenfall signed an agreement with Fermi Energia to become a minority shareholder of the company with an investment of €1 million ($1.2 million).

Fermi Energia has signed several agreements related to construction. In April 2022, Laurentis Energy Partners (a subsidiary of Canada's Ontario Power Generation) signed a cooperation agreement with Fermi Energia to develop a construction programme for SMRs in Estonia. In April 2025 South Korea's Samsung C&T Corporation Engineering & Construction Group signed a teaming agreement to collaborate on the deployment of two BWRX-300 units. Samsung C&T, with funding support from the South Korean government, subsequently launched a pre-feasibility study examining technical, financial, legal, and market aspects of SMR deployment, with a final report expected in 2026. Earlier in September 2025, Aecon signed an agreement with Fermi Energia to support development of a "capable reactor building deployment team" which would be "modelled after the Canadian experience"

referring to Aecon's work with Ontario Power Generation to construct a BWRX-300 at Darlington.

Site selection

In January 2025, Fermi Energia submitted an application to the country's Ministry of Economic Affairs and Communications to begin the planning process for its proposed 600 MWe two-unit BWRX-300 nuclear plant. The municipal councils of Viru-Nigula and Lüganuse formally agreed to participate.

In May 2025, the Estonian government formally initiated the national spatial planning process and strategic environmental impact assessment for the proposed plant. The planning area covers approximately 1285 square kilometres across multiple municipalities. Fermi Energia closed a €2.3 million funding round to finance the initial site pre-selection studies. The first part of the phased planning process, site pre-selection, is taking place between 2025 and 2027. The areas around the villages of Kunda in Viru-Nigula and of Aa in Lüganuse are the focus of studies, which look at the areas more broadly and consider various socioeconomic, environmental and infrastructure matters.

Assuming successful selection of a site, Fermi Energia plans to submit a construction permit application in 2029, with a view to operating the first of the two units from 2035.

Waste management

In November 2020, Deep Isolation, a radioactive waste disposal company, announced its collaboration with Fermi Energia on a preliminary study into using Deep Isolation's horizontal deep borehole disposal method in Estonia. Results from the study released in February 2021 concluded that there were no fundamental geological limitations to disposing of radioactive waste in deep horizontal boreholes in Estonia, and that such a repository could be used for the disposal of used fuel from the proposed SMRs.

Decommissioning

Estonia had two small Soviet naval reactors originally used for submarine training. They date from 1968 and 1983 and were

closed in 1989. They are in Safestor mode and will be dismantled after 50 years. The fuel has been returned to Russia.

Public opinion

Public views about the use of nuclear energy are generally positive.

Among Estonian respondents to the European Union's Special Eurobarometer 557 published in February 2025, 59% thought nuclear energy would have a positive effect on their way of life over the next two decades.

An earlier survey conducted in January 2022 by Kantar Emor on behalf of Fermi Energia found approximately 68% of Estonians either support or tend to support the consideration of small modular reactors (SMRs) in the country, an increase from 54% in January 2021.

Fermi Energia surveys in April 2025 showed national support at approximately 69%. In the areas potentially hosting the plant, local polling indicates around 50% are in favour and roughly one-third are opposed.

For more information see page on Nuclear Energy and Public Opinion.